SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project **R26-SE-009** | Agent 1 | **Nimthara Gunasinha** (IT22078582) | Reporting period **23.03.2026 – 25.08.2026**

1. PROJECT AND STUDENT DETAILS

Project ID	R26-SE-009
Research Topic	Self-Optimized LLM Multi-Agent System for Software Engineering
Supervisor	Prof. Nuwan Kodagoda
Co-Supervisor	Mr. Eishan Weerasinghe
Student Name	Nimthara Gunasinha
Student Registration No.	IT22078582
Individual Component	Agent 1 — Requirements Analyst (AI SRS Engineer) and the SRS user interface
Reporting Period	23 March 2026 to 25 August 2026
Phase Covered	Requirement, implementation and integration phase, representing approximately 65% of the planned project work.

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

This reporting period covers the development of Agent 1, the requirements and specification component of the multi-agent system, together with the SRS-facing user interface. The period began with a study of the SRS standards the generated document must satisfy and with the design of a guided intake flow that turns a plain project idea into structured interview questions, tracked answers, domain inference and a technology recommendation. The first implementation followed, covering session handling, question planning, specification generation and JSON and PDF artefact export, and was then given its own interface: intake forms, analysis screens, review tabs and the navigation handoff into the design-selection stage. An evaluation of that first version against the standard section list identified the gaps that shaped the rebuild, in which the component was restructured into independent intake, interview, planning, generation and document modules with schema-validated model output, a multi-source extraction pipeline, domain knowledge catalogues, authenticated endpoints and a plan runtime prepared for the builder handoff. Group-level contributions were made throughout to the shared AgentForge Studio workspace so that intake, review, artefacts and inter-agent handoff are demonstrable in one common interface.

3. DATE-WISE RP WORK (43 ENTRIES)

MARCH 2026

4 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
1	23.03.2026	Defined the scope of Agent 1 as the requirements and SRS analyst of the multi-agent system and identified the outputs the later agents depend on.

NO.	DATE	RP WORK CARRIED OUT
2	25.03.2026	Reviewed IEEE 830 and ISO/IEC/IEEE 29148 and fixed the section structure that every generated specification must follow.
3	27.03.2026	Studied how existing requirement-elicitation tools conduct structured interviews and recorded the techniques worth adopting.
4	30.03.2026	Agreed the Agent 1 to Agent 2 handoff with the group so the specification carries everything the builder stage needs.

APRIL 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
5	02.04.2026	Designed the guided intake flow that turns a plain project idea into a structured set of interview questions.
6	06.04.2026	Designed the section builders for purpose, scope, product functions, user classes, interfaces and non-functional requirement areas.
7	09.04.2026	Specified answer normalisation, open-item detection and completeness tracking so requirement gathering stays controlled and reviewable.
8	13.04.2026	Designed domain inference and the preliminary technology-stack recommendation carried through into the specification.
9	16.04.2026	Planned file-assisted intake so text, PDF, image and voice-note content can enrich requirement capture.
10	20.04.2026	Designed session persistence and artefact delivery so draft and final specifications can be saved, revisited and exported.
11	23.04.2026	Defined the validation rules, ambiguity reporting and readiness scoring applied before the formal handoff to Agent 2.
12	27.04.2026	Reviewed the complete intake-to-specification design against the supervisor's feedback and closed the remaining design points.

MAY 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
13	04.05.2026	Implemented the Agent 1 backend skeleton covering session handling, API wiring and the question-planning stage.
14	07.05.2026	Implemented specification generation for the core SRS sections and produced the first complete draft output.
15	11.05.2026	Added JSON and PDF artefact export so generated specifications can be reviewed outside the application.
16	14.05.2026	Planned the SRS front-end route structure and the screen breakdown for intake, analysis and review.

NO.	DATE	RP WORK CARRIED OUT
17	18.05.2026	Built the new-project intake forms with field validation for project brief, domain and constraint capture.
18	21.05.2026	Built the SRS screen tabs and the analysis flow that carries a project from intake through to the generated specification.
19	25.05.2026	Completed the styling assets and layout polish for the requirement analysis pages.
20	28.05.2026	Wired the SRS pages into the shared studio navigation and implemented the handoff into the design-selection stage.

JUNE 2026 8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
21	01.06.2026	Reviewed the first version output against the standard section list and recorded every gap to be closed in the rebuild.
22	04.06.2026	Tested the intake flow with sample project briefs and logged the cases where questioning failed to close requirement gaps.
23	08.06.2026	Studied structured-output techniques for model-driven specification writing so generated sections remain deterministic.
24	11.06.2026	Designed the module split so intake, interview, planning, generation and document output can evolve independently.
25	15.06.2026	Redesigned the interview stage so clarification questions are driven by measured coverage rather than a fixed question list.
26	18.06.2026	Prepared the interface reference screens for the redesigned SRS workspace and reviewed them with the group.
27	22.06.2026	Defined the project and specification schemas required for persistence in the rebuilt component.
28	25.06.2026	Specified the domain knowledge catalogue that drives domain classification and topic coverage.

JULY 2026 8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
29	01.07.2026	Rebuilt the SRS package on the new module structure and moved the service into the shared application layout.
30	06.07.2026	Implemented the structured model adapters so generation returns schema-validated output instead of free text.
31	09.07.2026	Implemented the project schemas and the repository layer for storing projects, intake data and generated specifications.
32	13.07.2026	Cleaned and consolidated the specification functions, removing superseded helpers and tightening the module boundaries.
33	16.07.2026	Built the source extraction pipeline for PDF, image and vision-derived requirement input.

NO.	DATE	RP WORK CARRIED OUT
34	20.07.2026	Extended extraction to speech-derived input and normalised every source into a single intake brief format.
35	23.07.2026	Aligned the specification request and response contract with the builder side and updated the shared package definitions.
36	28.07.2026	Tested the rebuilt pipeline against the earlier sample briefs and recorded the improvement in requirement coverage.

AUGUST 2026 7 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
37	03.08.2026	Implemented the domain classifier and the coverage auditor over the expanded knowledge catalogue.
38	06.08.2026	Implemented the clarification and interview flow that closes the gaps reported by the coverage auditor.
39	11.08.2026	Implemented customer-context capture so business background is carried through into the generated specification.
40	14.08.2026	Added access control and authentication to the plan and specification endpoints.
41	18.08.2026	Implemented plan assembly and the plan-merge logic that combines generated and clarified content.
42	21.08.2026	Strengthened the plan generation rules and reviewed the SRS workspace screens against the rebuilt backend flow.
43	25.08.2026	Verified the plan and specification output ahead of the builder handoff and consolidated the outcome of the reporting period.

4. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

1. Complete the diagram generation runtime and the professional diagram set produced alongside the specification.
2. Finalise the builder handoff contract so approved plans reach Agent 2 with full coverage and traceability data.
3. Raise document generation to the international SRS standard drafted for the component and validate the output against it.
4. Extend coverage auditing and clarification so requirement gaps are detected and closed before the specification is released.
5. Complete the SRS review, interview and diagram screens in the integrated studio and test them end to end with the other three components.

5. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

SUPERVISOR'S COMMENTS

Nimthara Gunasinha **STUDENT — IT22078582** Date: ______________	**Prof. Nuwan Kodagoda** **SUPERVISOR** Date: ______________	**Mr. Eishan Weerasinghe** **CO-SUPERVISOR** Date: ______________